Quiz one

- Time: 9:30-11:50
- Toilet breaks are not permitted during the exam.
- Calculators are NOT allowed.
- Write your answers on the provided answer sheets. Additional sheets will be supplied upon request. Ensure that your full name is clearly written on every page of your answer sheets.
- Show all computation steps in sufficient detail. Answers without supporting calculations will receive zero points, even if the final answer is correct.

Question 1. (4 marks) Find all vectors $\boldsymbol{a}$ that are orthogonal to the given vector $\boldsymbol{b}$ and have norm equal to the specified value d.

(i) (2 marks) $\boldsymbol{b} = \begin{pmatrix} 5, & 12 \end{pmatrix}$, $d = 26$

(ii) (2 marks) $\boldsymbol{b} = \begin{pmatrix} -6, & 8 \end{pmatrix}$, $d = 20$

Solution. Let $\boldsymbol{a} = \begin{pmatrix} a_1, & a_2 \end{pmatrix}$

1. By the given conditions, we have

$$\boldsymbol{a} \cdot \boldsymbol{b} = 5a_1 + 12a_2 = 0, \quad \|\boldsymbol{a}\| = \sqrt{a_1^2 + a_2^2} = 26.$$

Then we have the following two equations:

$$a_1 = -\frac{12}{5}a_2 \tag{1}$$
$$a_1^2 + a_2^2 = 26^2 \tag{2}$$

Substituting a_1 from Equation (1) into Equation (2), we obtain

$$a_2^2\left(\frac{144}{25} + 1\right) = 676 \Longrightarrow a_2^2 = 676 \times \frac{25}{169} = 100 \Longrightarrow a_2 = \pm 10 \quad \text{(1 mark)}$$

Together with Equation (1) we have

$$a_2 = 10 \Longrightarrow a_1 = -\frac{12}{5} \times 10 = -24, \quad a_2 = -10 \Longrightarrow a_1 = \frac{12}{5} \times 10 = 24.$$

Thus,

$$\boldsymbol{a} = \begin{pmatrix} -24, & 10 \end{pmatrix} \quad \text{or} \quad \boldsymbol{a} = \begin{pmatrix} 24, & -10 \end{pmatrix}. \quad \text{(1 mark)}$$

2. By the given conditions, we have

$$\boldsymbol{a} \cdot \boldsymbol{b} = -6a_1 + 8a_2 = 0, \quad \|\boldsymbol{a}\| = \sqrt{a_1^2 + a_2^2} = 20.$$

Then we have the following two equations:

$$a_1 = \frac{4}{3}a_2 \tag{3}$$
$$a_1^2 + a_2^2 = 400 \tag{4}$$

Substitute a_1 from Equation (3) into Equation (4) gives

$$a_2^2\left(\frac{16}{9}+1\right)=400 \Longrightarrow a_2^2=400\times\frac{9}{25}=144 \Longrightarrow a_2=\pm 12 \quad \text{(1 mark)}$$

Together with Equation (3) we have

$$a_2=12 \Longrightarrow a_1=16, \quad a_2=-12 \Longrightarrow a_1=-16 \quad \text{(1 mark)}$$

Thus

$$\boldsymbol{a}=\begin{pmatrix}16, & 12\end{pmatrix} \text{ or } \begin{pmatrix}-16, & -12\end{pmatrix}.$$

Question 2. (4 marks) Let

$$A=\begin{pmatrix}2 & -1 & 1\\ 0 & 3 & -2\\ 1 & 2 & 0\end{pmatrix}, \quad B=\begin{pmatrix}3 & 1 & 2\\ -2 & 4 & 0\\ 1 & -1 & 3\end{pmatrix}, \quad C=\begin{pmatrix}2 & 1 & 3\\ -1 & 4 & 0\\ 3 & -2 & 1\end{pmatrix}.$$

$$D=\begin{pmatrix}2 & 4 & 1 & -1\\ -1 & 3 & 2 & 0\\ 5 & 0 & -2 & 3\\ 1 & -2 & 4 & 2\end{pmatrix}, \quad E=\begin{pmatrix}4 & -1 & 2 & 0\\ 2 & 5 & 1 & -2\\ 3 & 2 & 4 & 1\\ -1 & 0 & 2 & 3\end{pmatrix}.$$

Compute the following matrix expressions.

(i) (2 marks) $A(BC)$

(ii) (2 marks) $2E^\top - 3D^\top$

Solution.

(i) First,

$$BC=\begin{pmatrix}3 & 1 & 2\\ -2 & 4 & 0\\ 1 & -1 & 3\end{pmatrix}\begin{pmatrix}2 & 1 & 3\\ -1 & 4 & 0\\ 3 & -2 & 1\end{pmatrix}$$

$$=\begin{pmatrix}3\times 2+1\times(-1)+2\times 3 & 3\times 1+1\times 4+2\times(-2) & 3\times 3+1\times 0+2\times 1\\ (-2)\times 2+4\times(-1)+0\times 3 & (-2)\times 1+4\times 4+0\times(-2) & (-2)\times 3+4\times 0+0\times 1\\ 1\times 2+(-1)\times(-1)+3\times 3 & 1\times 1+(-1)\times 4+3\times(-2) & 1\times 3+(-1)\times 0+3\times 1\end{pmatrix}$$

$$=\begin{pmatrix}11 & 3 & 11\\ -8 & 14 & -6\\ 12 & -9 & 6\end{pmatrix} \text{(1 mark)}$$

Therefore,

$$A(BC)=\begin{pmatrix}2 & -1 & 1\\ 0 & 3 & -2\\ 1 & 2 & 0\end{pmatrix}\begin{pmatrix}11 & 3 & 11\\ -8 & 14 & -6\\ 12 & -9 & 6\end{pmatrix}$$

$$=\begin{pmatrix}2\times 11+(-1)\times(-8)+1\times 12 & 2\times 3+(-1)\times 14+1\times(-9) & 2\times 11+(-1)\times(-6)+1\times 6\\ 0\times 11+3\times(-8)+(-2)\times 12 & 0\times 3+3\times 14+(-2)\times(-9) & 0\times 11+3\times(-6)+(-2)\times 6\\ 1\times 11+2\times(-8)+0\times 12 & 1\times 3+2\times 14+0\times(-9) & 1\times 11+2\times(-6)+0\times 6\end{pmatrix}$$

$$=\begin{pmatrix}42 & -17 & 34\\ -48 & 60 & -30\\ -5 & 31 & -1\end{pmatrix} \text{(1 mark)}$$

(ii)
$$E^\top = \begin{pmatrix} 4 & 2 & 3 & -1 \\ -1 & 5 & 2 & 0 \\ 2 & 1 & 4 & 2 \\ 0 & -2 & 1 & 3 \end{pmatrix}, \qquad D^\top = \begin{pmatrix} 2 & -1 & 5 & 1 \\ 4 & 3 & 0 & -2 \\ 1 & 2 & -2 & 4 \\ -1 & 0 & 3 & 2 \end{pmatrix} \quad \text{(1 mark)}$$

Therefore,
$$2E^\top - 3D^\top = 2\begin{pmatrix} 4 & 2 & 3 & -1 \\ -1 & 5 & 2 & 0 \\ 2 & 1 & 4 & 2 \\ 0 & -2 & 1 & 3 \end{pmatrix} - 3\begin{pmatrix} 2 & -1 & 5 & 1 \\ 4 & 3 & 0 & -2 \\ 1 & 2 & -2 & 4 \\ -1 & 0 & 3 & 2 \end{pmatrix}$$
$$= \begin{pmatrix} 8 & 4 & 6 & -2 \\ -2 & 10 & 4 & 0 \\ 4 & 2 & 8 & 4 \\ 0 & -4 & 2 & 6 \end{pmatrix} - \begin{pmatrix} 6 & -3 & 15 & 3 \\ 12 & 9 & 0 & -6 \\ 3 & 6 & -6 & 12 \\ -3 & 0 & 9 & 6 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 7 & -9 & -5 \\ -14 & 1 & 4 & 6 \\ 1 & -4 & 14 & -8 \\ 3 & -4 & -7 & 0 \end{pmatrix} \quad \text{(1 mark)}$$

- Incorrect transpose: (−0.5 marks).
- Each incorrect entry in the final matrix results in (−0.1 marks).

Question 3. (7 marks) Consider the following system of linear equations:
$$\begin{aligned} x_1 + 2x_2 - x_3 &= a, \\ 2x_1 + 3x_2 - 2x_3 &= b, \\ -x_1 - x_2 + x_3 + x_4 &= c, \\ 2x_1 + 4x_2 - 2x_3 + x_4 &= d. \end{aligned}$$

(i) (0.5 marks) Write the augmented matrix corresponding to the given system of linear equations.

(ii) (2 marks) Apply Gauss–Jordan elimination to the augmented matrix until the coefficient matrix is in reduced row echelon form.

(iii) (0.5 marks) Determine the condition on a, b, c, d for which this linear system is *inconsistent*.

(iv) (4 marks) If the system is consistent, determine whether it has a unique solution or infinitely many solutions. Express the solution(s) in terms of a, b, c, d.

Solution.

(i) The augmented matrix corresponding to the system is
$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 2 & 3 & -2 & 0 & b \\ -1 & -1 & 1 & 1 & c \\ 2 & 4 & -2 & 1 & d \end{array}\right) \quad \text{(0.5 marks)}$$

(ii) Applying Gauss–Jordan elimination, we have

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 2 & 3 & -2 & 0 & b \\ -1 & -1 & 1 & 1 & c \\ 2 & 4 & -2 & 1 & d \end{array}\right) \xrightarrow[R_3\to R_3+R_1,\ R_4\to R_4-2R_1]{R_2\to R_2-2R_1} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 0 & -1 & 0 & 0 & b-2a \\ 0 & 1 & 0 & 1 & c+a \\ 0 & 0 & 0 & 1 & d-2a \end{array}\right)$$

$$\xrightarrow{R_2\to -R_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 0 & 1 & 0 & 0 & 2a-b \\ 0 & 1 & 0 & 1 & c+a \\ 0 & 0 & 0 & 1 & d-2a \end{array}\right) \xrightarrow{R_3\to R_3-R_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 0 & 1 & 0 & 0 & 2a-b \\ 0 & 0 & 0 & 1 & b+c-a \\ 0 & 0 & 0 & 1 & d-2a \end{array}\right)$$

$$\xrightarrow{R_4\to R_4-R_3} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 0 & a \\ 0 & 1 & 0 & 0 & 2a-b \\ 0 & 0 & 0 & 1 & b+c-a \\ 0 & 0 & 0 & 0 & d-a-b-c \end{array}\right)$$

$$\xrightarrow{R_1\to R_1-2R_2} \left(\begin{array}{cccc|c} 1 & 0 & -1 & 0 & -3a+2b \\ 0 & 1 & 0 & 0 & 2a-b \\ 0 & 0 & 0 & 1 & b+c-a \\ 0 & 0 & 0 & 0 & d-a-b-c \end{array}\right) \quad \text{(2 marks)}$$

- Not applying Gauss–Jordan elimination or not completing the required reduction: (−0.5 marks).
- Not showing intermediate matrices: (−0.5 marks).
- Missing or incorrect row-operation labels: up to (−0.5 marks).
- Each independent arithmetic error: (−0.25 marks). Do not penalize propagated errors repeatedly.
- Accept alternative valid sequences of elementary row operations.

(iii) If

$$d-a-b-c \neq 0,$$

then the last row represents an equation of the form

$$0 = \text{nonzero constant}$$

Therefore, the system is inconsistent if

$$d-a-b-c \neq 0 \quad \text{(0.5 marks)}$$

(iv) Suppose that

$$d = a+b+c.$$

Then the reduced system is

$$\begin{aligned} x_1 - x_3 &= -3a+2b, \\ x_2 &= 2a-b, \\ x_4 &= b+c-a. \quad \text{(2 marks)} \end{aligned}$$

x_3 is a free variable. Let

$$x_3 = t, \qquad t \in \mathbb{R}.$$

Then

$$\begin{aligned} x_1 &= -3a + 2b + t, \\ x_2 &= 2a - b, \\ x_3 &= t, \\ x_4 &= b + c - a. \end{aligned}$$

Therefore, the system has infinitely many solutions:

$$\{(-3a + 2b + t,\ 2a - b,\ t,\ b + c - a) \mid t \in \mathbb{R}\}. \quad \text{(2 marks)}$$

There is no choice of a, b, c, d for which the system has a unique solution, since x_3 is a free variable whenever the system is consistent.

Question 4. (10 marks) Let

$$A = \begin{pmatrix} -3 & 1 \\ 2 & 2 \end{pmatrix}$$

(i) (2 marks) Use the inversion algorithm to find A^{-1}. That is, adjoin I_2 to the right of A and apply elementary row operations to reduce A to I_2.

(ii) (2 marks) Write down the elementary matrix corresponding to each elementary row operation used in part (i).

(iii) (5 marks) Find the inverses of the elementary matrices from part (ii), and hence express A as a product of elementary matrices.

(iv) (1 mark) Express A^{-1} as a product of elementary matrices.

Solution.

(i) (2 marks) We apply the inversion algorithm to find the inverse of A and record the elementary matrices corresponding to each row operation during the process

$$\left(\begin{array}{cc|cc} -3 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \to -\frac{1}{3}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 2 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & \frac{8}{3} & \frac{2}{3} & 1 \end{array}\right) \xrightarrow{R_2 \to \frac{3}{8}R_2}$$

$$\left(\begin{array}{cc|cc} 1 & -\frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{4} & \frac{3}{8} \end{array}\right) \xrightarrow{R_1 \to R_1 + \frac{1}{3}R_2} \left(\begin{array}{cc|cc} 1 & 0 & -\frac{1}{4} & \frac{1}{8} \\ 0 & 1 & \frac{1}{4} & \frac{3}{8} \end{array}\right)$$

We have

$$A^{-1} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{3}{8} \end{pmatrix}$$

(ii) (2 marks) The four matrices are

$$E_1 = \begin{pmatrix} -\frac{1}{3} & 0 \\ 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 \\ 0 & \frac{3}{8} \end{pmatrix}, \quad E_4 = \begin{pmatrix} 1 & \frac{1}{3} \\ 0 & 1 \end{pmatrix}$$

(iii) The inverses of the elementary matrices are

$$E_4^{-1} = \begin{pmatrix} 1 & -\frac{1}{3} \\ 0 & 1 \end{pmatrix}, \quad E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{8}{3} \end{pmatrix}, \quad E_2^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad E_1^{-1} = \begin{pmatrix} -3 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{(4 marks)}$$

We have

$$E_4E_3E_2E_1A = I \Longrightarrow A = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}. \quad \text{(1 mark)}$$

(iv) Consequently

$$A^{-1} = E_4E_3E_2E_1. \quad \text{(1 mark)}$$

- Part (i): Not using the specified inversion algorithm: $(-0.5$ marks).
- Part (i): Missing intermediate matrices: $(-0.5$ marks).
- Part (i): Missing or incorrect row-operation labels: up to $(-0.5$ marks).
- Part (i): Each independent arithmetic error: $(-0.25$ marks).
- Part (iii): Each correct inverse elementary matrix earns 1 mark; the correct factorization earns 1 mark.
- Part (iv): Correct product expression earns 1 mark.